

$$\Gamma(-1) = (-2)! = \infty$$

$$\Gamma(0) = (-1)! = \infty$$

$$\Gamma(1) = 0! = 1$$

$$\Gamma(2) = 1! = 1$$

$$\Gamma(3) = 2! = 2$$

$$\Gamma(4) = 3! = 6$$

$$\Gamma(-\frac{3}{2}) = \frac{4}{3}\sqrt{\pi} \approx 2.363\,271\,801\,207$$

$$\Gamma(-\frac{1}{2}) = -2\sqrt{\pi} \approx -3.544\,907\,701\,811$$

$$\Gamma(\frac{1}{2}) = \sqrt{\pi} \approx 1.772\,453\,850\,905$$

$$\Gamma(\frac{3}{2}) = \frac{1}{2}\sqrt{\pi} \approx 0.886\,226\,925\,45$$

$$\Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi} \approx 1.329\,340\,388\,18$$

$$\Gamma(\frac{7}{2}) = \frac{15}{8}\sqrt{\pi} \approx 3.323\,350\,970\,45$$